

From Anxiety to Assurance: A Continuous Multi-Physics Model with 5G RRC Dynamics

Summary

To address "battery anxiety," this paper formulates a **continuous-time system** governed by coupled Ordinary Differential Equations to predict Time-to-Empty (TTE). We construct an integrated "**Electrical-Thermal-Aging**" **coupled framework** to describe the dynamic behavior of Lithium-ion batteries. The electrical subsystem employs a **second-order RC equivalent circuit model** combined with the **Ampere-hour integral method**. Recognizing the bidirectional influence between temperature and electrical parameters, we simultaneously establish a **thermodynamic equilibrium equation** based on Newton's Law of Cooling. This mechanism dynamically updates internal resistance and capacity parameters via polynomial fitting surfaces, reflecting the complex interaction where electrical discharge generates heat, and heat, in turn, alters electrical efficiency. Additionally, the **Arrhenius equation** and a **power-law aging model** are introduced to correct maximum capacity based on cycle number and temperature, ensuring lifecycle validity.

To address intermittent network power consumption, we innovatively incorporate a **5G Radio Resource Control (RRC) state machine model**. This maps discrete communication states to continuous power fluctuations, capturing the "tail energy" often overlooked by static models. We defined **six typical usage scenarios** by decomposing total load into component-level models. Simulation results indicate the model accurately predicts SOC decay, aligning well with empirical data. The analysis reveals that high screen brightness and CPU load are the primary drivers of "avalanche-style" power drops in active use.

Finally, we performed a One-at-a-Time (OAT) sensitivity analysis to assess the robustness of the predictions. The analysis results show that the **environmental temperature** is the most significant non-linear factor; while in daily usage, the **screen brightness** and **CPU utilization** are the key factors affecting TTE. Based on these quantitative findings, we recommend that users utilize auto-brightness features to mitigate exponential screen power consumption and that app developers employ batch data transmission strategies to reduce the frequency of RRC state transitions and the associated tail energy waste. In conclusion, this continuous-time multi-physics model provides a robust tool for quantifying battery behavior, offering theoretical support for alleviating user anxiety and optimizing system scheduling strategies.

Keywords: Continuous-Time Model; Second-order RC Circuit; Thermal-Electrical Coupling; RRC State Machine; Time-to-Empty Prediction.

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1 Introduction

1.1 Background

Smartphones have been deeply integrated into the fabric of modern society, becoming the central medium for information exchange, entertainment, and work. However, with the exponential growth in processor performance, the widespread adoption of 5G communication technology, and the iteration of screen display technologies, the contradiction between the surging energy demands of mobile devices and the slow growth of battery energy density has become increasingly prominent. For users, unexpected battery depletion dynamics not only cause inconvenience but may also lead to serious consequences in emergency situations. This "range anxiety" has become a widespread socio-psychological phenomenon.

In reality, the discharge behavior of a battery is the combined result of various complex factors. Under different usage scenarios and patterns, power consumption often exhibits drastically different profiles. Furthermore, the degree of battery aging further exacerbates the difficulty of prediction. Therefore, establishing a physically grounded, continuous-time battery charge prediction model for smartphones is of significant theoretical importance and practical value for understanding power consumption mechanisms, optimizing system scheduling, and alleviating user anxiety.



Figure 1: The anxiety about low battery levels among people in modern society^[13]

1.2 Restatement of the Problem

This project requires us to develop a continuous-time mathematical model for a lithium-ion battery-powered smartphone that can accurately predict the battery's State of Charge (SOC) over time under realistic usage conditions. Based on this model, we analyze the battery depletion time under different usage scenarios. The specific tasks can be summarized as follows:

- (1) Establish a continuous-time multi-physics coupled model: Develop a system of differential equations that describes the battery SOC as a function of time, and consider

continuously extending the model to incorporate additional factors such as the screen, CPU, and network modules.

- (2) Multi-scenario prediction and validation: Using the constructed model, predict the remaining battery discharge time under given different operating conditions. Compare simulation results with actual discharge data or empirical values, analyze the model's applicability under different load characteristics, and quantify the key drivers leading to rapid battery depletion.
- (3) Sensitivity and robustness analysis: Appropriately change modeling assumptions, model parameters, or usage patterns to perform a sensitivity analysis on the model. Investigate the response of model outputs to parameter perturbations, thereby testing the model's sensitivity and robustness.
- (4) Strategy recommendations and model generalization: Based on the quantitative conclusions derived from the model, provide specific battery-saving behavior recommendations for users, and discuss the model's strengths, weaknesses, and generalization potential.

1.3 Our Work

To address the above problems, this paper constructs an integrated "thermal-electrical-aging" coupled prediction framework. Our workflow mainly consists of the following four stages:

- First, establish an electrochemical model based on a second-order RC circuit. We abandon the simple ideal battery assumption and adopt a second-order RC equivalent circuit to simulate the ohmic internal resistance, electrochemical polarization, and concentration polarization phenomena within the lithium-ion battery. Using the Ampere-hour integral method as the main line for SOC calculation, we incorporate the Arrhenius equation to introduce aging corrections to the battery's maximum capacity Q_n based on cycle count and temperature.
- Second, introduce the thermal-electrical coupling mechanism and the RRC state machine. Considering the significant impact of temperature on battery internal resistance and capacity, we establish a battery thermal equilibrium equation, achieving real-time bidirectional coupling between battery temperature and electrical parameters. Simultaneously, targeting the intermittent data transmission characteristics in 5G communication scenarios, we innovatively introduce a Radio Resource Control (RRC) state machine model, mapping discrete communication states to continuous power fluctuations, thereby more accurately capturing network standby power consumption.
- Then, conduct multi-scenario simulations and driving factor analysis. We performed full-cycle discharge simulations under various typical scenarios including standby, video playback, high-load gaming, and dynamic communication. By comparing the

SOC decline curves and temperature rise curves under different operating conditions, we validated the model's accuracy and identified that high screen brightness and high CPU load are the primary causes of "avalanche-style" power drops, while frequent network state transitions are the main culprit of hidden power consumption.

- Finally, perform sensitivity analysis and strategy formulation. We analyzed the model's sensitivity to usage scenarios and patterns. Based on the analysis results, we formulated scientific battery-saving guidelines for users and indicated potential future extensions of the model.

2 Assumptions and Justification

Considering that practical problems always contain many complex factors, first of all, we need to make reasonable assumptions to simplify the model, and each hypothesis is closely followed by its corresponding explanation:

- **Assumption 1: Uniform temperature distribution within the battery.**

Smartphone batteries are typically flat pouch cells with relatively thin thickness, and the thermal conductivity of their internal materials is relatively high. Compared to the natural convection heat exchange between the battery and the environment, the heat conduction speed inside the battery is sufficiently fast. Therefore, the internal temperature gradient can be neglected, and the battery can be treated as a lumped thermal mass system with uniform temperature, following Newton's Law of Cooling.

- **Assumption 2: The electrochemical characteristics of the battery can be approximated by a second-order RC equivalent circuit.**

Although the internal processes of a battery involve complex lithium-ion diffusion and charge transfer, a second-order RC model consisting of one series resistor and two parallel RC branches has been widely proven in engineering to accurately simulate the transient response of batteries. One RC branch models the charge transfer process, and the other models the diffusion process, achieving an optimal balance between computational efficiency and physical realism.

- **Assumption 3: Circuit parameters depend solely on SOC and temperature.**

The battery's open-circuit voltage U_{ocv} , ohmic internal resistance R_0 , and polarization parameters R_p , C_p are primarily influenced by the current electrochemical state and reaction activity. Ignoring hysteresis effects and other minor factors, treating these parameters as binary functions of SOC and temperature is a standard and effective approach.

- **Assumption 4: Power consumption of different components satisfies the principle of linear superposition.**

In practical systems, each module operates largely independently under the scheduling of the operating system, and energy consumption mainly originates from the active state of hardware modules. Therefore, this assumption is reasonable in most scenarios.

- **Assumption 5: Ambient temperature remains constant during a single discharge process.**

Phone discharge typically occurs in a relatively stable environment (e.g., indoors or in a pocket). Compared to the battery's own temperature rise, the surrounding environment acts as a large thermal reservoir with minimal temperature variation. Therefore, assuming the ambient temperature T_a as a constant is a reasonable boundary condition.

- **Assumption 6: The timescale of battery aging is much longer than a single discharge process.**

Battery capacity fading is caused by long-term cycling leading to SEI layer thickening and active lithium loss, which typically become apparent only after hundreds of cycles. During the few hours of a single discharge, the battery's maximum capacity Q_n is treated as a constant. Its aging effects are only reflected in the correction of Q_n using different cycle counts (N) as an initial condition.

- **Assumption 7: The 5G RRC state machine can be simplified to three states: IDLE, INACTIVE, CONNECTED, and state transitions are controlled solely by data packet arrival events and timers.**

This abstraction retains the key low-power states and tail-energy characteristics of the protocol, effectively reflecting the energy consumption dynamics under intermittent data transmission.

3 Notations

Some important mathematical notations used in this paper are listed in Table 1.

Table 1: Notations

Symbol	Meaning	Dimension	Symbol	Meaning	Dimension
$SOC(t)$	State of Charge (SOC)	%	M	Battery mass	kg
Q_n	Actual battery capacity considering aging and temperature	Ah	A	Battery surface area for heat dissipation	m ²
$Q_{n,ref}$	Maximum capacity at reference temperature	Ah	h	Convective heat transfer coefficient	W/(m ² ·K)
N	Battery cycle count	-	P_{total}	Total smartphone power consumption	W
E_{act}	Activation energy	eV	P_{screen}	Screen power consumption	W
a	Capacity fade coefficient	-	P_{CPU}	Total processor power consumption	W
α	Capacity fade exponent	-	P_{work}	CPU full-load power consumption	W
R	Gas constant	eV/K	P_{idle}	CPU idle power consumption	W
U_0/U_{ocv}	Battery open-circuit voltage	V	u	CPU utilization	%
U	Battery terminal voltage	V	P_{GPS}	GPS module power consumption	W
I	Load current (discharge positive)	A	β_{GPS}	Average power in GPS active state	W
U_{p1}, U_{p2}	Polarization voltages of RC branches	V	u_{GPS}	Ratio of GPS active time	-
R_0	Ohmic internal resistance	Ω	$P_{background}$	Background service baseline power consumption	W
R_{p1}	Electrochemical polarization resistance	Ω	$P_{network}$	Network module power consumption	W
R_{p2}	Concentration polarization resistance	Ω	$\beta_{network}$	Energy coefficient for network data transmission	W/pps
C_{p1}	Electrochemical polarization capacitance	F	β_{TRX}	Radio frequency baseline power consumption	W
C_{p2}	Concentration polarization capacitance	F	u_{TRX}	Packet transmission rate	pps
T	Battery core temperature	K	T_1	RRC inactivity timer	s
T_{ref}	Reference temperature	K	T_2	RRC deep sleep timer	s
T_a	Ambient temperature	K	γ	Transmission energy efficiency coefficient	W/Mbps
P_{gen}	Battery heat generation power	W			
C	Battery specific heat capacity	J/(kg·K)			

4 Continuous-Time Smartphone Battery Depletion Prediction Model

To accurately predict the Time-to-Empty of a smartphone battery under complex operating conditions, this study establishes an integrated "electrical-thermal-aging" multi-physics coupled model. This model is not merely a fit to discrete data but is a set of

continuous-time differential equations constructed based on the internal physicochemical mechanisms of lithium-ion batteries. The model consists of three subsystems: a second-order RC equivalent circuit model describing the battery's electrical characteristics, a thermal equilibrium model describing temperature dynamics, and an RRC state machine load model describing intermittent communication characteristics.

4.1 Second-Order RC Equivalent Circuit Model Based on the Ampere-Hour Integral Method

According to Kirchhoff's Voltage Law (KVL), the battery terminal voltage $U(t)$ can be expressed as the difference between the open-circuit voltage and internal voltage drops:

$$U(t) = U_{ocv}(SOC) - I(t)R_0 - U_{p1}(t) - U_{p2}(t) \quad (1)$$

Where U_{ocv} is the battery open-circuit voltage (OCV); $I(t)$ is the load current (positive for discharge); R_0 is the ohmic internal resistance, primarily reflecting the instantaneous impedance of the electrolyte and electrode materials. To accurately capture the charge transfer polarization at the electrode surface and the concentration polarization within the active materials, this paper introduces two parallel RC branches, used to simulate the fast charge transfer process and double-layer effect, and the slower lithium-ion diffusion process, respectively.

Based on circuit principles, the dynamic changes of the polarization voltages U_{p1} and U_{p2} are described by the following differential equations:

$$\begin{cases} \frac{dU_{p1}}{dt} = \frac{I(t)}{C_{p1}} - \frac{U_{p1}}{R_{p1}C_{p1}} \\ \frac{dU_{p2}}{dt} = \frac{I(t)}{C_{p2}} - \frac{U_{p2}}{R_{p2}C_{p2}} \end{cases} \quad (2)$$

The above equations constitute the second-order RC equivalent circuit model, which can continuously describe the transient response of the battery's internal potential to load changes.

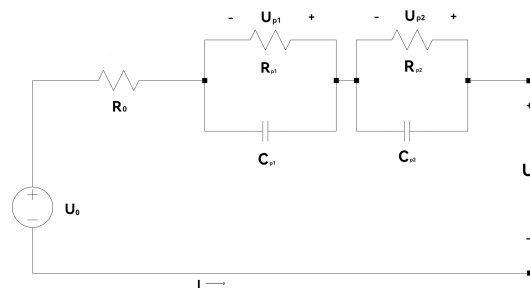


Figure 2: Schematic diagram of the second-order RC equivalent circuit model

The traditional Ampere-hour Integral method often assumes a constant total battery capacity, which can lead to significant errors over long periods or under extreme temperatures. This model introduces a dynamic actual capacity $Q_n(N, T)$, correcting it to be a function of the battery core temperature T and the battery cycle count N .

According to the Arrhenius equation and a power-law aging model^{[7][2]}, the actual battery capacity considering aging and temperature is expressed as:

$$Q_n(N, T) = Q_{n,ref} \cdot \left[1 - \underbrace{a \cdot N^\alpha}_{\text{Cycle Aging}} - \underbrace{\exp\left(\frac{E_{act}}{R} \left(\frac{1}{T_{ref}} - \frac{1}{T(t)}\right)\right)}_{\text{Temperature Effect}} \right] \quad (3)$$

Where $T_{ref} = 298.15 \text{ K}$ is the reference temperature, $Q_{n,ref}$ is the maximum capacity at the reference temperature, $E_{act} = 0.4 \text{ eV}$ is the activation energy, $a = 1.5 \times 10^{-4}$ is the capacity fade coefficient, $\alpha = 0.6$ is the capacity fade exponent, $R = 8.617 \times 10^{-5} \text{ eV/K}$ is the Boltzmann constant.

Thus, the rate of change of the battery state of charge ($SOC(t)$) is defined as:

$$\frac{dSOC(t)}{dt} = -\frac{I(t)}{Q_n(N, T(t))} \quad (4)$$

Where Q_n is the total battery capacity. This equation indicates that the capacity Q_n degradation caused by high temperature or multiple cycles will accelerate the decline rate of $SOC(t)$, thereby shortening the battery life.

So far, the second-order RC equivalent circuit model based on the Ampere-hour integral method has been established. By determining all circuit parameters through offline or online identification, the battery's charge consumption state over time can be obtained.

4.2 Model Extension for Thermal-Electrical Coupling

Battery discharge is often accompanied by significant thermal effects, and temperature changes, in turn, alter the battery's internal resistance and capacity, forming a bidirectional coupling mechanism of "electricity generates heat, and heat in turn alters electrical properties." To expand the model's applicability, we consider constructing a thermal-electrical coupling mechanism to extend our base model.

The battery's heat generation power P_{gen} mainly consists of Joule heat (irreversible heat) and polarization heat. According to energy conservation, the heat generation power can be calculated as the product of the current and the total overpotential:

$$P_{gen}(t) = I(t) \cdot (U_{ocv} - U) = I(t)^2 R_0 + I(t)U_{p1} + I(t)U_{p2} \quad (5)$$

Assuming a uniform temperature distribution inside the battery, it is treated as a lumped thermal mass system. According to the first law of thermodynamics and New-

ton's Law of Cooling, the evolution of the battery core temperature $T(t)$ follows the following differential equation:

$$C \cdot M \frac{dT}{dt} = P_{gen}(t) - h \cdot A \cdot (T(t) - T_a) \quad (6)$$

Where C is the battery specific heat capacity; for the LiFePO_4 battery of interest in this paper, $C = 1.022 \text{ J/(g} \cdot \text{K)}$; M is the battery mass; T is the battery temperature; T_a is the ambient temperature, assumed to be room temperature 300 K under general usage conditions; $P_{gen} = I(U - U_{ocv})$ is the battery heat generation power; A is the battery surface area for heat dissipation; h is the convective heat transfer coefficient, taken as $h = 12 \text{ W/(m}^2 \cdot \text{K)}$ for a Li battery in contact with air.

This equation can describe the change in battery temperature due to heat generation during discharge and heat exchange with the external environment. In summary, this paper establishes a comprehensive thermal-electrical coupling mechanism: during smartphone battery use, a certain current I causes the battery to generate heat and change its temperature. The temperature, on one hand, in turn changes the relationship between SOC and circuit parameters, thereby altering the current I through the equivalent circuit model; on the other hand, it also changes the battery's maximum capacity at the current temperature, which in turn changes the measured SOC value through the Ampere-hour integral. This coupling mechanism can better simulate real-world usage scenarios and provide more accurate battery depletion prediction results.

4.3 Method for Determining Equivalent Circuit Parameters

The circuit parameters U_{ocv} , R_0 , R_{p1} , C_{p1} , R_{p2} , C_{p2} in the model all vary with SOC and temperature. To obtain their continuous functional forms, we performed surface fitting using bivariate quadratic polynomials based on discrete parameter data at different temperature and SOC points provided in the literature^[5], resulting in the continuous temperature-SOC relationships for each circuit parameter:

$$\begin{cases} R_0 = 0.011105 + 0.000479SOC + 0.000012T + 0.000001SOC - 0.000006T - 0.000001SOC \cdot T \\ R_1 = 0.053828 - 0.002316SOC + 0.000641T + 0.000035SOC + 0.000051T - 0.000032SOC \cdot T \\ R_2 = 0.053828 - 0.002316SOC + 0.000641T + 0.000035SOC + 0.000051T - 0.000032SOC \cdot T \\ C_1 = 35.199206 - 0.679908SOC + 0.108700T + 0.003568SOC - 0.002257T - 0.000747SOC \cdot T \\ C_2 = 2695.603541 - 37.650371SOC + 10.446465T + 0.123904SOC - 0.113732T - 0.053350SOC \cdot T \end{cases}$$

The literature^[1] presents a classic method for obtaining the relationship curve between battery open-circuit voltage U_{ocv} and SOC: constant current discharge, pausing for 1 minute at fixed capacity intervals, and recording the terminal voltage during the pause (Rapid test). After obtaining the data, a sixth-degree polynomial curve fit is applied to the U_{ocv} -SOC data. According to literature^[6], we obtained the functional expression for the battery open-circuit voltage U_{ocv} with respect to SOC:

$$U_{ocv} = 11.08 \cdot SOC^6 - 25.58 \cdot SOC^5 + 17.54 \cdot SOC^4 - 1.159 \cdot SOC^3 - 2.386 \cdot SOC^2 + 1.263 \cdot SOC + 3.422$$

Through the above fitting, all circuit parameters are expressed as continuous functions of SOC and temperature, allowing the differential equation system to update parameters in real-time during numerical solution, thereby more accurately simulating the battery's dynamic response.

4.4 Analysis Method for Loads under Different Operating Conditions

Regarding the load, we consider the power P of each component under different operating conditions and establish component power models.

Given the total power consumption $P_{total}(t) = U(t) \cdot I(t)$, combining it with the terminal voltage equation yields a quadratic equation in terms of current $I(t)$:

$$P_{total} = I \cdot (U_{net} - IR_0) \implies R_0 I^2 - U_{net} I + P_{total} = 0$$

where $U_{net} = U_{ocv} - U_{p1} - U_{p2}$ is the net electromotive force after deducting polarization voltages. According to the quadratic formula, the physically reasonable discharge current solution (taking the smaller root to conform to steady-state) is:

$$I(t) = \frac{U_{net} - \sqrt{U_{net}^2 - 4R_0 P_{total}}}{2R_0} \quad (7)$$

The total power consumption of a smartphone originates from multiple hardware components such as the screen, processor, communication module, GPS, and background services. To simulate real user scenarios, we decompose the total smartphone power consumption P_{total} into a linear superposition of individual hardware modules:

$$P_{total}(t) = P_{screen} + P_{CPU} + P_{GPS} + P_{network} + P_{background} \quad (8)$$

The power consumption models for each component are established based on measured data, with the modeling rationale as follows:

1. P_{screen}

Screen power primarily depends on brightness and often exhibits an exponential relationship. By fitting an exponential function to discrete data points from literature^[3], we obtained:

$$P_{screen} = 0.0076554 \cdot e^{0.016 \cdot Bright} \quad (9)$$

2. P_{CPU}

According to literature^[4], processor power has a linear relationship with utilization (u):

$$P_{CPU} = (P_{work} - P_{idle}) \cdot u + P_{idle} \quad (10)$$

where P_{work} is the full-load power, set to 6 W for an 8-core processor; P_{idle} is the idle power, set to a typical value of 50 mW; u is the utilization, within $[0\%, 100\%]$, set according to typical application scenarios. According to test documentation from the battery testing website^[11], in typical gaming scenario battery discharge tests, CPU utilization is usually moderate.

3. P_{GPS}

According to literature^[4], P_{GPS} can be calculated using:

$$P_{GPS} = \beta_{GPS} \cdot u_{GPS} \quad (11)$$

where β_{GPS} is the Active state average power, taken as 250 mW; u_{GPS} is the ratio of active time per unit time (Ratio of active time).

4. $P_{network}$

According to literature^[4], under WLAN connection, $P_{network}$ and u_{TRX} exhibit a piecewise linear relationship.

$$P_{WiFi} = \begin{cases} \beta_{LT} \times u_{TRX} + \beta_{LT_{TRX}}, & \text{if } u_{TRX} \leq \text{Threshold} \\ \beta_{HT} \times u_{TRX} + \beta_{HT_{TRX}}, & \text{if } u_{TRX} > \text{Threshold} \end{cases} \quad (12)$$

However, the threshold is extremely low, approximately 20 pps. Therefore, for simplicity, we assume $P_{network}$ and u_{TRX} always satisfy the second segment relationship.

$$P_{network} = \beta_{network} \cdot u_{TRX} + \beta_{TRX} \quad (13)$$

where $\beta_{network} = 0.5$ is the coefficient related to data rate, $\beta_{TRX} = 200$ mW, u_{TRX} is in pps (packets per second).

5. $P_{background}$

Background services, based on daily experience and set to a constant 10 mW.

6. Power Table of Components

Based on the above models, we defined five typical usage scenarios (Standby, Communication, Video Playback, Gaming, Navigation) and their corresponding component power parameters, summarized in the table below. These scenarios cover typical power consumption patterns from low to high load, providing input conditions for subsequent simulations.

Table 2: Power Table of Components

Scenario	P_{screen}	P_{CPU}	P_{GPS}	$P_{network}$	$P_{background}$	P_{total}
(Standby)	0	50	0	50	10	110
(Call/Chat)	75	645	25	207.5	10	962
		($u = 0.1$)	($u_{GPS} = 0.1$)	($u_{TRX} = 15$)		
(Video)	75	942.5	0	300	10	1327.5
		($u = 0.15$)		($u_{TRX} = 200$)		
(Game)	414	3025	0	260	10	3709
		($u = 0.5$)		($u_{TRX} = 120$)		
(Nav)	414	1240	200	250	10	2114
		($u = 0.2$)	($u_{GPS} = 0.8$)	($u_{TRX} = 100$)		

4.5 Further Extension: RRC State Machine Model for Intermittent Data Transmission

The communication module of a smartphone, especially the mobile cellular network interface, is one of the primary contributors to system power consumption. Its power consumption characteristics are not constant but are closely related to the network activity state. When the device is transmitting data, components such as the modem and RF front-end are in a high-power active state. During periods of no data transmission, to quickly respond to potential network requests and avoid the latency and signaling overhead associated with frequent connection establishment, modern mobile communication protocols introduce complex Radio Resource Control (RRC) state machine management mechanisms.

To enable our prediction model to better describe the process of intermittent data transmission, this paper constructs a simplified three-state RRC dynamic system model to simulate the power consumption dynamics of a smartphone under a cellular network driven by intermittent data traffic. The core idea of this model is to abstract the energy consumption of the network interface into three discrete states (`IDLE`, `INACTIVE`, `CONNECTED`) and conditional transitions between them. Each state corresponds to a baseline power level, and in the `CONNECTED` state, a dynamic power term related to the instantaneous data rate is introduced.

State Definitions and Transition Logic

The RRC state machine defined in this model includes the following three states:

- **RRC_IDLE:** In this state, the device maintains only minimal synchronization with the network and has no dedicated wireless resource channel. Power consumption is the lowest. However, to initiate or receive data transmission, a full signaling procedure is required to switch to the `CONNECTED` state, introducing a delay on the order of tens to hundreds of milliseconds.
- **RRC_INACTIVE:** This is an energy-saving state introduced in 5G NR and similar to states in later 4G evolution. In this state, the device retains part of the core network context but releases air interface resources. Power consumption is slightly higher than in the `IDLE` state, but recovery to the `CONNECTED` state is much faster than waking from `IDLE`, achieving a better balance between latency and power consumption^[8].
- **RRC_CONNECTED:** The device has a dedicated control channel and possibly a data channel, enabling high-speed, low-latency data transmission. This state has the highest power consumption and can be further subdivided into an active transmission period and a tail waiting period.

Transitions between states are triggered by two key events: data transmission requests and timer expiration. When the application layer has data packets to send or receive, regardless of whether the current state is `IDLE` or `INACTIVE`, the device immediately

transitions to the `CONNECTED` state to perform data transmission. After data transmission stops, the device does not immediately return to a low-power state but starts timers. There are two timers: the Inactivity Timer ($T1$) and the Deep Sleep Timer ($T2$). Timing starts from the end of the last data transmission. If no new data transmission occurs within the duration $T1$, the state drops from `CONNECTED` to `INACTIVE`. The existence of $T1$ leads to "tail energy consumption," meaning that after data transmission is complete, the device maintains a higher power level for a period to prepare for rapid subsequent data transmission. After entering the `INACTIVE` state, another timer $T2$ is started. If no data transmission occurs within the duration $T2$, the state further drops to the lowest-power `IDLE` state.

The following pseudo-code illustrates the transition logic of the state machine:

Algorithm 1 RRC State Machine Transition Logic

```

1: Initialization:  $\text{current\_state} \leftarrow \text{IDLE}$ ,  $\text{timer\_T1} \leftarrow 0$ ,  $\text{timer\_T2} \leftarrow 0$ 
2: for each time step  $t$  do
3:   Get current data rate  $R_{bps}(t)$ 
4:   if  $R_{bps}(t) > 0$  then
5:      $\text{current\_state} \leftarrow \text{CONNECTED}$ 
6:      $\text{timer\_T1} \leftarrow 0$ ,  $\text{timer\_T2} \leftarrow 0$  ▷ Reset all timers
7:   else if  $R_{bps}(t) == 0$  then
8:     if  $\text{current\_state} == \text{CONNECTED}$  then
9:        $\text{timer\_T1} \leftarrow \text{timer\_T1} + \Delta t$ 
10:      if  $\text{timer\_T1} \geq T1$  then
11:         $\text{current\_state} \leftarrow \text{INACTIVE}$ 
12:         $\text{timer\_T2} \leftarrow 0$  ▷ Enter INACTIVE state, reset T2 timer
13:      end if
14:    else if  $\text{current\_state} == \text{INACTIVE}$  then
15:       $\text{timer\_T2} \leftarrow \text{timer\_T2} + \Delta t$ 
16:      if  $\text{timer\_T2} \geq T2$  then
17:         $\text{current\_state} \leftarrow \text{IDLE}$ 
18:      end if
19:    end if
20:  end if
21: end for

```

State-Power Mapping

Each RRC state is mapped to a specific power consumption value P_{state} , while the power consumption of the `CONNECTED` state is additionally proportional to the instantaneous data rate R_{bps} .

The power consumption of the `IDLE` state, P_{IDLE} , is the baseline power required for the device to maintain minimal synchronization with the network, typically low. Based on measurement data from literature^[8], this paper sets $P_{IDLE} = 0.11$ W.

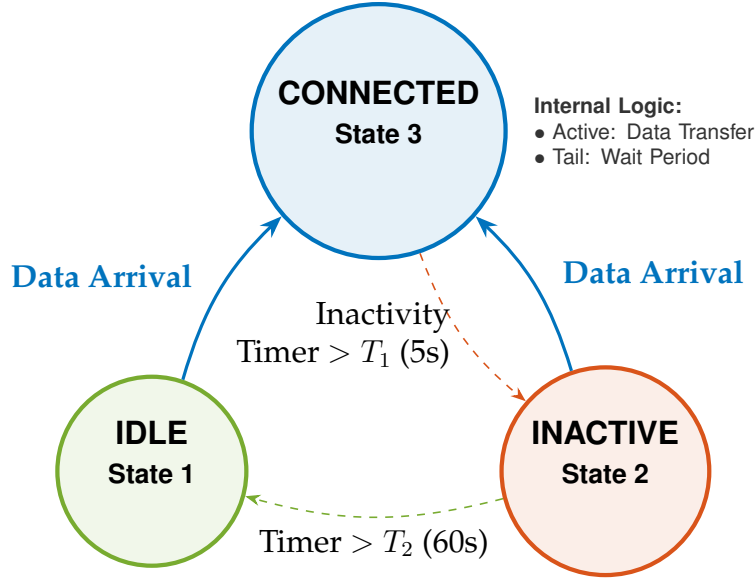


Figure 3: RRC State Transition Logic and Power Consumption Diagram

The power consumption of the INACTIVE state, $P_{INACTIVE}$, is slightly higher than that of the IDLE state, as the device needs to maintain more context information for rapid recovery. According to reference [8], we set $P_{INACTIVE} = 0.15 \text{ W}$.

The power consumption of the CONNECTED state is more complex and consists of two parts. First, the baseline connection power: when in the CONNECTED state but without data transmission (i.e., the tail period), the device still occupies radio resources, with a power consumption of P_{CONN_TAIL} . Combining references [8] and [10], this paper sets $P_{CONN_TAIL} = 0.8 \text{ W}$.

Second, the additional power consumption for data transmission: when there is data transmission ($R_{bps} > 0$), the power consumption is the sum of the baseline power P_{CONN_ACTIVE} and an increment proportional to the data rate R_{bps} . That is:

$$P_{CONNECTED}(R_{bps}) = P_{CONN_ACTIVE} + \gamma \cdot R_{bps} \quad (14)$$

Here, P_{CONN_ACTIVE} is the baseline connection power at zero data rate, set to 1.2 W in this paper; γ is the transmission energy efficiency coefficient, representing the additional power consumption per Mbps data rate. According to reference [9], this paper sets $\gamma = 0.5 \text{ W/Mbps}$.

Dynamic Load Generation and Model Integration

To simulate typical intermittent communication scenarios, such as social applications periodically fetching new messages, this paper defines a periodic data transmission pattern: a data pulse with a duration of 0.5 seconds and a rate of 5 Mbps is generated every 30 seconds, simulating a brief data exchange. There is no data transmission during the remaining time.

Using this data transmission pattern as input to drive the aforementioned RRC state machine model and power mapping relationships, we can obtain a dynamically changing power consumption curve $P_{comm}(t)$ for the communication module. This dynamic power function $P_{comm}(t)$ is treated as an independent usage scenario for 5G RRC dynamic communication. By feeding it into the unified battery differential equation system, we can observe and analyze the stepwise decay of the battery state of charge and the tiny 'ripple'-like fluctuations in battery temperature under intermittent, pulsed network activity.

By introducing this model, the battery discharge prediction framework in this paper can more realistically reflect the energy consumption of smartphones in typical fragmented usage. This provides a quantitative analysis tool for users to understand 'standby power consumption' and for operating systems to design smarter network state scheduling strategies. The model also indicates that optimizing application data transmission patterns, such as batch transmission to reduce state transitions, may lead to significant battery life improvements.

4.6 Simulation Results

Based on the previously constructed "thermal-electrical-aging" coupled second-order RC circuit model and the RRC state machine model for intermittent communication, we substituted the identified physical parameters into the system and conducted full-cycle discharge simulations for six typical operating conditions: standby, constant communication, video playback, gaming, navigation, and 5G RRC dynamic communication. We employed the Fourth-order Runge-Kutta method (RK4) for numerical integration. Figures 4 to 8 demonstrate the model's predictive performance across different dimensions and its comparative analysis with real-world data.

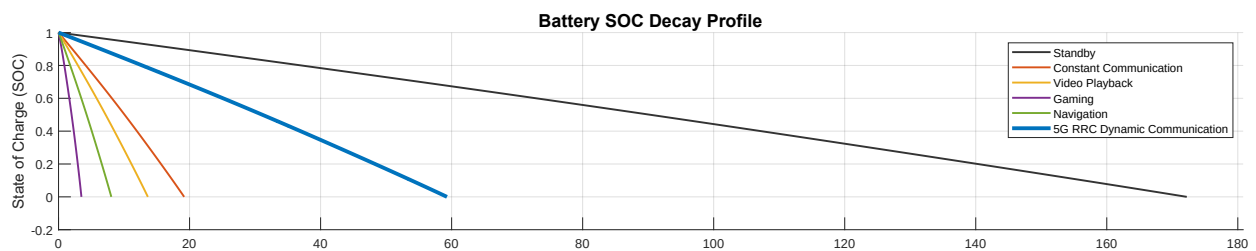


Figure 4: SOC Depletion Prediction under Different Usage Scenarios

Figures 4 and 5 reveal the significant impact of different loads on the battery SOC and core temperature. Simulation results indicate that the power consumption rates across scenarios exhibit distinct hierarchical differences: the standby mode has the longest battery life; however, after introducing intermittent data transmission (5G RRC Dynamic), despite the low average data rate, the battery life is significantly reduced to about one-third compared to pure standby, limited by the "tail power" effect in the RRC protocol. In high-load scenarios, Gaming and Navigation not only show the steepest SOC decline slopes but are also accompanied by the most significant temperature rises. Notably, the

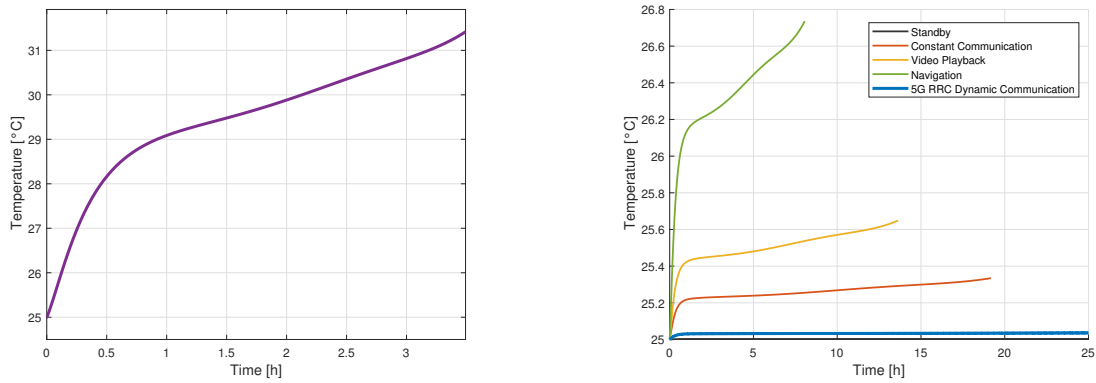


Figure 5: Battery Temperature Prediction under Different Usage Scenarios

power consumption of the Navigation scenario even exceeds that of continuous video playback, which is mainly attributed to the combined effect of frequent GPS module calls and constant screen illumination. Simultaneously, the temperature curves and SOC decline curves show a strong correlated coupling characteristic: the more intense the discharge scenario, the faster the temperature rise due to the Joule heating effect, and the high temperature further exacerbates internal resistance losses, validating the effectiveness of the "electrical-thermal" coupling mechanism in the model.

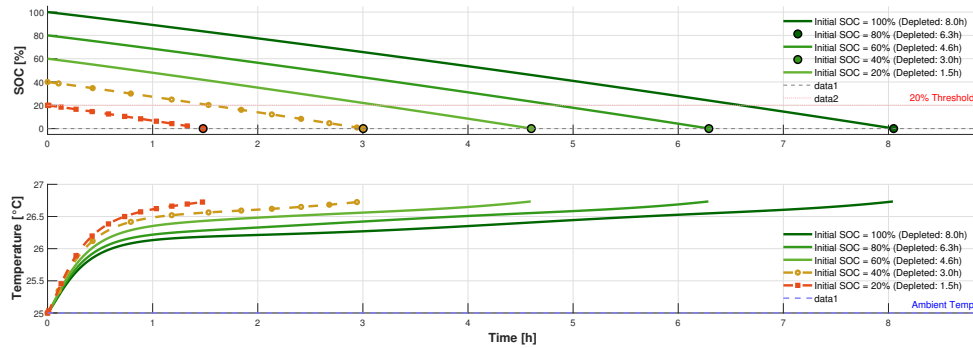


Figure 6: SOC Depletion and Temperature Prediction under Different Initial Charge Levels

Figure 6 shows the discharge trajectories under different initial SOC. The results indicate that the decline slopes of the curves remain highly consistent under the same load, demonstrating that the battery discharge behavior has good linear independence from the initial charge level. This means the model can extrapolate the battery life performance for any remaining charge level through a single full-cycle simulation.

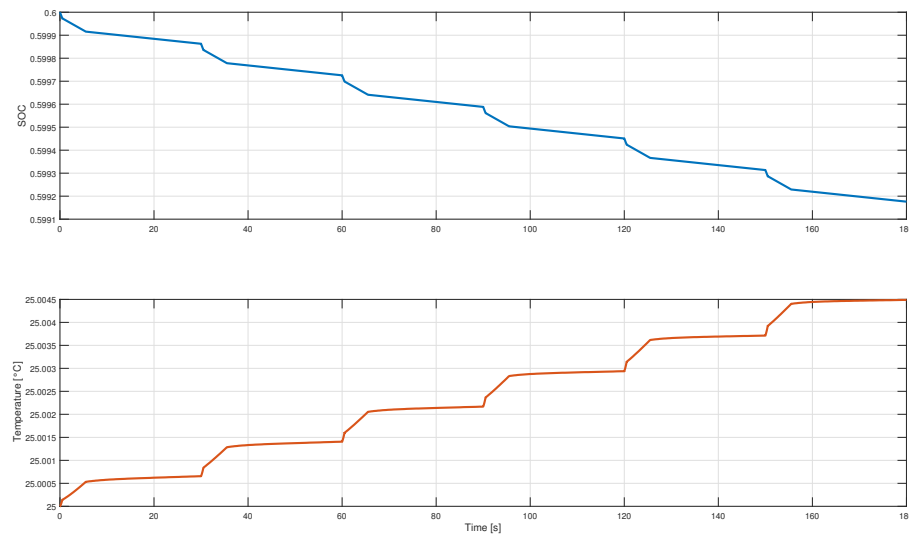


Figure 7: Instantaneous SOC and Temperature Variations under the RRC State Machine Model

Figure 7, from a microscopic perspective, shows the transient response under the RRC state machine model: the SOC exhibits a characteristic "stepwise" decline, accompanied by pulsed, minor fluctuations in temperature (Thermal Ripples). This detail accurately captures the instantaneous high-energy consumption characteristics triggered by packet arrivals causing state machine transitions, proving the superiority of this sub-model in describing fragmented network activities.

To test the model's real-world applicability, we compared the simulation results with measured data from the gsmarena^[11] and dxomark^[12] websites.

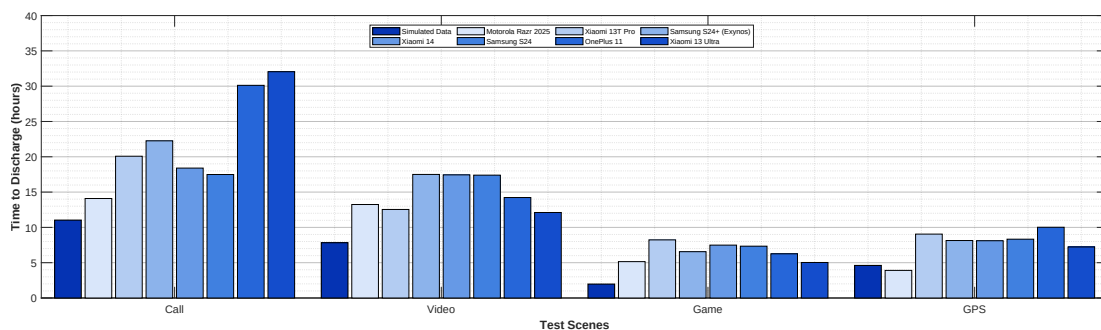


Figure 8: Comparison between Simulated Prediction Data and Real Mobile Phone Discharge Data

The model's predicted values for video playback and GPS navigation scenarios show high agreement with the measured averages of multiple phone models, proving the effectiveness of the component power superposition principle. However, there are some

deviations between the model's predictions and some phone models in constant communication and gaming scenarios. This may be due to different manufacturers employing aggressive dynamic voltage and frequency scaling strategies or differentiated thermal designs in specific high-load scenarios, causing actual power consumption to deviate from the theoretical model.

Based on the conclusions, this paper analyzes the specific driving factors for rapid battery depletion in each case. In the 5G dynamic communication scenario, the primary driving force for battery depletion is not the data transmission itself but stems from the tail effect of the Radio Resource Control protocol. For the video playback scenario, the exponential power consumption of the screen component is the dominant factor. In contrast, the gaming and navigation scenarios exhibit more complex multi-module concurrency and thermal-electrical coupling driving characteristics. The rapid power drop in the gaming scenario is driven by the combination of high computational load and the resulting thermal effects, while the driving factor for power drop in the navigation scenario lies in the full-load operation of all hardware modules.

5 Sensitivity Analysis

To evaluate the robustness of the model and identify the key drivers leading to rapid battery depletion, we need to examine the sensitivity of the model's output to changes in input parameters. Considering the uncertainties present in real-world usage, this paper employs the One-at-a-Time (OAT) method for univariate local sensitivity analysis, separately studying the impact of user behavior parameters and physical environmental parameters on battery life prediction.

5.1 Analysis Method and Parameter Selection

This paper sets the baseline scenario as the typical "Navigation Condition" (high screen brightness, GPS on, medium CPU load). Centered around this baseline, we perturb the selected key parameters within a range of $\pm 25\%$, with a step size of 5% , resulting in a total of 11 test points. At each test point, keeping other parameters unchanged, we re-run the battery discharge simulation and record the change in the battery's time-to-empty.

Based on the model structure, we selected the following six representative parameters for analysis: screen brightness index L_{br} , CPU utilization u , GPS active time ratio u_{GPS} , network data transmission rate u_{TRX} , ambient temperature T_a , and battery cycle count N . We define the quantitative indicator of sensitivity as:

$$S_i = \frac{\Delta t_{\text{empty}}}{\Delta p_i} \times \frac{p_{i,\text{base}}}{t_{\text{base}}} \quad (15)$$

5.2 Analysis Results

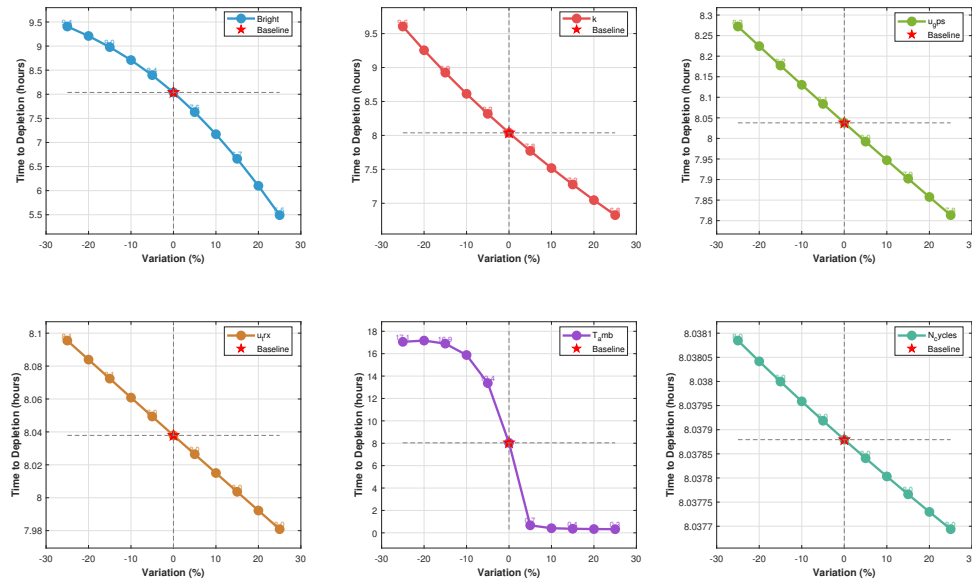


Figure 9: Sensitivity Analysis of Six Parameters

Simulation results indicate that temperature has a nonlinear and decisive influence on battery life, while the impact of cycling aging in the short term is negligible.

Ambient temperature is the most sensitive parameter in the model, exhibiting extreme nonlinear characteristics: lithium-ion batteries perform optimally near room temperature (20-30°C), and any deviation from this range significantly affects battery life. When the ambient temperature increases by only 5% (from 298.15 K to 313.06 K), the simulated battery life plummets from 8.04 hours to 0.67 hours; when the temperature increases by 25% (to 372.69 K), the battery life further shortens to 0.32 hours. This indicates that the Arrhenius capacity correction term in the model is highly sensitive to high temperatures, simulating the instantaneous "thermal collapse" of effective capacity caused by intensified side reactions at high temperatures. Conversely, when the temperature decreases by 25% (to 223.61 K), the model predicts a discharge time extending to 17.06 hours. This reflects that under low load power, low temperatures significantly suppress the capacity decay term, increasing the theoretical available capacity calculated by the model. The fluctuation range of T_{end} caused by temperature changes is as high as 16.74 hours, far exceeding all other factors.

Under the baseline condition of a new battery with $N = 1$, perturbing the cycle count within the range $[0.75, 1.25]$ results in T_{end} consistently stabilizing at 8.04 hours. This shows that in the early stages of battery life, the capacity loss calculated by the power-law aging model ($a \cdot N^\alpha$) is extremely small, and its impact on a single discharge process is negligible.

After excluding extreme data, the influence of other user behavior parameters on battery life shows a clear hierarchical difference.

Screen brightness and CPU utilization are the core variables determining daily battery life, both showing a significant negative correlation with battery life: when the brightness index increases by 25% from 250 to 312.5, the power consumption rises to 2.83 W, and the battery life shortens from 8.04 hours to 5.49 hours (a decrease of 31.7%). Conversely, reducing brightness by 25% can extend battery life to 9.41 hours. Its influence range is about 4 hours, making it the most influential among user-controllable parameters. When CPU utilization increases by 25%, battery life decreases to 6.83 hours; reducing it by 25% increases battery life to 9.60 hours. Its influence range is about 2.8 hours, second only to screen brightness. This verifies that "screen-on usage" and "high-computation tasks" are the two main drivers causing rapid battery drain.

In contrast, perturbations in the GPS and network communication modules have a relatively limited impact on overall battery life: with $\pm 25\%$ fluctuation in the GPS active ratio (u_{GPS}), the battery life only fluctuates slightly between 7.81h and 8.27h, with a change range less than 0.5 hours. With $\pm 25\%$ fluctuation in the network rate (u_{trx}), the battery life remains stable between 7.98h and 8.10h, with a change range of only 0.12 hours. This indicates that in high baseline noise scenarios like navigation, the power consumption influence of GPS and network is overwhelmed by the main power consumers and no longer constitutes the primary contradiction.

5.3 Summary and Discussion

Based on the simulation results, we calculated the local sensitivity coefficients of each factor within a variation range of $\pm 5\%$. The $\pm 5\%$ range was chosen because the response of each factor is approximately linear within this range, corresponding to common variation magnitudes in actual use.

Table 3: Calculated Sensitivity Coefficients for Each Factor ($\pm 5\%$ Variation Range)

Influencing Factor	Baseline Value	Time-to-Empty at -5% (h)	Time-to-Empty at +5% (h)	Sensitivity Coefficient S
Ambient Temperature (T_{amb})	298.15 K	13.37	0.67	-15.80
Screen Brightness (Bright)	250	8.40	7.63	-0.96
CPU Utilization (k)	0.2	8.32	7.77	-0.68
GPS Usage Rate (u_{GPS})	0.8	8.08	7.99	-0.11
Network Transmission Rate (u_{trx})	100	8.05	8.03	-0.025
Battery Cycle Count (N_{cycles})	1	8.04	8.04	0.00

It should be noted that this sensitivity analysis uses the equal percentage change method ($\pm 25\%$) centered on each parameter's baseline value. Although this method is widely used in engineering analysis for an intuitive comparison of the relative influence of different factors, its fairness and practical guidance have certain limitations for parameters with vastly different orders of magnitude in their baseline values. This limitation is mainly reflected in the two extremes of ambient temperature and cycle count.

For ambient temperature T_a , its baseline value is 298.15 K on the thermodynamic temperature scale. Due to the large base of the Kelvin scale, a relative change of $\pm 25\%$ corresponds to a substantial absolute physical increment: the high-temperature side reaches

373 K, and the low-temperature side reaches 223 K. This magnitude of perturbation actually exceeds the comfort zone for normal smartphone use, crossing into the realm of extreme environments. Therefore, the extremely high sensitivity exhibited by temperature is partly because the perturbation covers the physical boundaries of battery failure.

Conversely, for battery cycle count N , the baseline is set to $N = 1$. At this baseline, even a 25% increase only changes N to 1.25. For a lithium-ion battery with a typical lifespan of 500-1000 cycles, the change from the 1st to the 1.25th cycle is negligible in terms of physicochemical properties. This percentage perturbation based on the state of a new device masks the true influence of aging factors over long-term use.

Through data-based sensitivity analysis, the following clear conclusions can be drawn: Ambient temperature achieves a high sensitivity coefficient in the simulation. Under extreme temperature conditions, the battery life of a mobile phone will be dominated by ambient temperature. However, under normal temperature conditions, screen brightness and CPU load are the key factors affecting battery life. The use of network communication and the activation of GPS have a relatively limited impact on battery life.

6 Recommendations for Smartphone Users

Based on our "thermal-electrical-aging" coupled model and sensitivity analysis results, we find that battery depletion is not determined by a single factor alone, but rather by the combined effects of screen power consumption, processor load, ambient temperature, and network communication mechanisms. To alleviate users' "battery anxiety" and extend device usage time, we propose the following scientifically grounded recommendations based on the physical model:

First, screen brightness management is the most direct and effective means for users to extend battery life. Since screen power consumption has an exponential relationship with brightness, it is recommended that users enable the "auto-brightness" feature in daily use to avoid manually maintaining unnecessarily high brightness indoors; when the battery is critically low, manually reducing the brightness to the minimum visible level can significantly slow down the SOC decline rate. Second, users should be highly vigilant about the hidden damage caused by the "electrical-thermal" vicious cycle to the battery. The model shows that high temperatures significantly increase battery internal resistance and accelerate capacity degradation. Therefore, behaviors that cause device overheating, such as "playing while charging," should be avoided. Simultaneously, try to avoid prolonged use of the phone in extreme temperature environments, such as inside a car exposed to summer sun or in extremely cold outdoors, to keep the battery operating within its optimal electrochemical activity range.

Furthermore, regarding the hidden power consumption of the communication module, users should pay attention to their network connection habits. Given the tail power consumption inherent in the RRC state machine, frequent and fragmented data interactions can cause the communication module to remain in a high-power state for extended periods. It is recommended that users or developers utilize the system's batch transmission mechanism to consolidate message pushes and background synchronization, reduc-

ing frequent state machine wake-ups. Finally, the cumulative effect of silent loads should not be overlooked. Although the power consumption of a single background process is tiny, the integral effect over a long period is significant. Users should develop the habit of regularly clearing high-energy-consuming background applications or use system permission management to block unnecessary apps from auto-starting and silent wake-up paths, thereby plugging the "tiny leaks" of battery power at the source.

7 Strengths and Weaknesses

Strengths

(1) Deep Multi-Physics Coupling Mechanism

This model abandons the traditional single-circuit model and innovatively establishes a closed-loop "electrical-thermal-aging" coupled system. By modeling the circuit parameters of the second-order RC circuit as bivariate continuous functions of temperature T and state of charge SOC , and incorporating Newton's Law of Cooling to provide real-time feedback on temperature changes, the model successfully captures the voltage drop phenomenon caused by temperature rise in high-load scenarios. Its physical realism far exceeds that of static models.

(2) Accurate Characterization of Microscopic Communication Energy Consumption

Addressing the flaw of traditional models often treating network power consumption as a constant, this paper introduces an RRC state machine model. This sub-model can accurately simulate state transitions and "tail power consumption" caused by intermittent packet transmission in 5G communication, effectively explaining why phones still exhibit noticeable power consumption in low-traffic standby states. It enhances the model's prediction accuracy in dynamic network scenarios.

(3) Continuous Model Based on Actual Physical Mechanisms

Unlike black-box models based on curve fitting, our system of differential equations explicitly reveals the dynamic feedback mechanisms between current, heat, and internal resistance. This allows us not only to predict when the battery drains but also to explain why distinct "avalanche" depletion occurs in specific scenarios.

Weaknesses

(1) Limitations of the Lumped Thermal Capacitance Assumption

In the thermal model, this paper assumes a uniform internal temperature distribution within the battery, treating it as a lumped thermal capacitance system. However, in real-world high-power fast charging or extreme discharge scenarios, significant thermal gradients exist inside the battery, potentially leading to localized overheating. Neglecting the spatial dimension of heat distribution may cause deviations in the model when predicting extreme thermal runaway conditions.

(2) Generalization Error Due to Hardware Heterogeneity

During the validation phase comparing seven different phone models, the model exhibited some fluctuations. This is because the model employs a universal parameter set that does not fully encompass the specific Battery Management System strategies of different manufacturers. The model currently lacks adaptive adjustment capabilities for hardware differences across specific device models.

(3) Short-Term Approximation of Battery Aging Cycle

Although the model considers the correction of capacity by cycle count N , the impact of aging is treated as a static initial condition in simulations of a single discharge process. For full lifecycle predictions spanning several years, the model does not yet include the dynamic evolution of complex electrochemical aging mechanisms, such as electrolyte drying or lithium plating.

8 Future Work

To further enhance the model's prediction accuracy and generalizability, we plan to deepen research in the following directions:

1. Establishment of a Spatially Distributed Thermal Model

We will upgrade the lumped thermal model to two-dimensional or three-dimensional heat conduction equations. Using finite element analysis methods, we will simulate the heat flow distribution inside the battery pack and between the battery, the phone's motherboard, and the casing. This will help identify local hot spots, not only enabling more accurate prediction of average temperature but also providing a theoretical basis for optimizing the design of the phone's thermal dissipation structure.

2. Tracking Full Lifecycle Aging Trajectories

Extend the model's time scale from a single discharge process to the battery's entire lifecycle. We will introduce more complex electrochemical aging mechanisms and study the long-term cumulative impact of user charging habits on *SOH*, thereby providing more forward-looking battery maintenance recommendations.

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9 Report on Use of AI Tools

1. Google Gemini (Feb 1, 2026 version, Gemini 3)

Query: <Among the collected literature, the second-order RC equivalent circuit model of lithium-ion batteries is mainly used to analyze the battery state of power storage devices. Can this model be applied to predict the changes in the state of charge (SOC) of mobile phone batteries?>

Output: <Absolutely. The second-order RC equivalent circuit model (2RC ECM) is not only applicable to power batteries (e.g., in electric vehicles) but also suitable for lithium batteries in consumer electronics like smartphones. Here is a detailed analysis:

1. Physical Universality The electrochemical characteristics of lithium-ion batteries are fundamentally universal. Regardless of battery size, the same physicochemical processes occur internally: Ohm's internal resistance: Resistance of the electrolyte and tabs. Electrochemical polarization (first-order RC): Reaction kinetics process. Concentration polarization (second-order RC): Diffusion process within active materials. The second-order model can more accurately capture the high-frequency voltage fluctuations and recovery characteristics during frequent "wake-sleep" switching in phones compared to the first-order model.

2. Advantages for Phone SOC Prediction In smartphone application scenarios, using the second-order RC model offers the following benefits: Dimensional performance accuracy: Phone loads fluctuate greatly (e.g., sudden launch of a large app). The second-order model better simulates the voltage response under such dynamics, reducing SOC estimation errors. Computational load: Although more complex than the first-order model, modern smartphone processors (SoCs) have far greater computing power than automotive Battery Management Systems (BMS), making second-order matrix operations trivial. Application environment: Phones have higher requirements for charging time prediction (e.g., "20 minutes until full"). The second-order model provides more accurate polarization simulation during the final charging stage.

3. Practical Application Limitations While the model is universal, differences exist in parameter identification between phone batteries and power batteries: Rate differences: Phones typically do not involve high-rate discharges (e.g., hundreds of amps) like power batteries. Environmental changes: Phones are primarily used at ambient temperatures, while power batteries require consideration of stringent temperature control. Therefore, for phone applications, temperature compensation parameters can be appropriately simplified.

Conclusion: The second-order RC model is a classic choice balancing "computational accuracy" and "real-time performance." If you are developing a SOC estimation algorithm for mobile devices (e.g., combined with Extended Kalman Filter EKF), the second-order RC model is an ideal physical model foundation.

Would you like me to provide the mathematical state-space representation of this model for your simulation experiments?>

2. Google Gemini (Feb 1, 2026 version, Gemini 3)

Query: <Help me find papers that study the power consumption of various components of smart phones (such as screen usage, processor load, network connection, GPS usage)>

Output: <Regarding research on power consumption of smartphone components, the academic community has accumulated a very substantial body of literature. This type of research is often termed "Power Modeling" or "Energy Profiling."

Here are several milestone or core papers that cover all the components you mentioned (screen, CPU, network, GPS), categorized by research perspective:

1. Foundational Classic Papers: Full Component Power Breakdown If you need to understand what percentage of power each internal hardware component consumes, these two are essential reading:

An Analysis of Power Consumption in a Smartphone (Aaron Carroll & Gernot Heiser, USENIX ATC)

Core content: One of the most highly cited papers in this field. The researchers physically "surgerized" a phone, installing resistors on each module (CPU, RAM, screen, GPU, GSM, WiFi, GPS, audio, etc.) for measurement.

Conclusions: Provides detailed power consumption percentages for each component under different usage scenarios (e.g., standby, call, web browsing). For example, screen backlight and cellular network are often the biggest "power hogs."

Understanding and Reducing Smartphone Energy Consumption (Aaron Carroll PhD thesis / related papers)

Core content: Deeply dissects the power consumption of modern multi-core smartphones, with detailed experiments particularly on the nonlinear relationship between screen brightness, processor load & frequency (DVFS).

2. Software Modeling and Load Prediction: How Components Interact If you are more concerned with how "load" affects power consumption, the following research provides mathematical models:

PowerBooster: Automated Power Model Generation for Wireless Mobile Devices (Zhang et al., MobiSys)

Core content: Proposes the PowerBooster model. It doesn't require disassembling the phone but infers total power consumption by monitoring system states (e.g., CPU usage, screen brightness level, packet transmission rate, GPS satellite connection count).

Component analysis: Provides classic descriptions of GPS's "asynchronous power consumption" (i.e., residual power effects after being turned off) and the state machine of network connection (RRC state transitions).

Accurate CPU Power Modeling for Multicore Smartphones (Microsoft Research)

Core content: Specifically focuses on processor load. Studies the complex impact of activating different numbers of cores and changing frequencies on power consumption in multi-core architectures, correcting errors in earlier simple linear models.

3. In-Depth Studies on Specific Components Regarding Screen (Display): * *The Impact of Display Type on Smartphone Energy Consumption*: Compares the power consumption differences between LCD and OLED, noting that OLED power consumption is highly dependent on the displayed content's colors (pixel self-emission).

Regarding Network & GPS (Network & Location): * *Characterizing Power Consumption of Dual-Frequency GNSS of Smartphone* (arXiv): Analyzes the additional power overhead of dual-frequency GPS compared to single-frequency GPS during satellite acquisition and tracking phases.

Energy Consumption of 4G and 5G Interfaces: Compares the tail power consumption (Tail Energy) under different network connection modes.

Search Suggestions You can further search using the following keywords in Google Scholar or IEEE Xplore:

Smartphone Energy Profiling

Power Breakdown of Mobile Devices

Power modeling of GPS/Display/CPU in Android

Would you like me to delve deeper into the specific power calculation formulas or influencing factors for a particular component (e.g., "screen" or "GPS")?>

3.Google Gemini (Feb 1, 2026 version, Gemini 3)

Query: <How many pps of data are transmitted per second when chatting purely in text using Telegram?>

Output: <Here is a detailed analysis of packets per second (pps) for Telegram's pure text chat:

1. Instant of Message Sending (Burst Traffic) When you click the "send" button, a brief interaction occurs at the network layer: Uplink (Client -> Server): Generates 2-4 packets. Contains the encrypted message payload (Encrypted Message). MTProto protocol-level sequence numbers and acknowledgment requests. Downlink (Server -> Client): Generates 2-3 packets. Server receipt acknowledgment (ACK). Status updates (e.g., message successfully synced to the cloud, triggering the logic for single tick to double tick). Instantaneous pps: Within tens of milliseconds of sending, pps spikes instantly. Averaged over a full second, sending one text message generates approximately 5-10 pps.

2. Average Value During Continuous Conversation During a normal text conversation frequency (e.g., a few messages per minute): Average pps: Stabilizes around 1-2 pps. Reason: Telegram uses a long-lived connection (Long-polling or Push mechanism). If both parties are in the chat window, there are minimal status synchronization packets (e.g., "typing..."), which are typically very small and have fixed frequencies.

3. Telegram's Protocol Characteristics Unlike Facebook, etc., which use standard HTTP-S/TLS, Telegram's MTPROTO protocol has slight differences in pps performance: Heartbeat packets (Keep-alive): To maintain background connections on mobile devices, Telegram sends a heartbeat packet approximately once per minute. If not chatting, pps is almost 0.02. MTU utilization: Pure text packets are often much smaller than the Ethernet MTU (1500 bytes). The encrypted packet size for an ordinary short sentence is typically between 150-300 Bytes.

Summary Comparison Scenario Estimated pps (Packets Per Second) Notes Completely Silent / Background < 0.1 pps Only very low-frequency heartbeat packets to maintain connection Continuously Sending Text 2 - 5 pps Includes "typing" status and ACK acknowledgments Click Send Instant 8 - 15 pps Extremely brief handshake and acknowledgment burst Expert Perspective If you are doing battery drain analysis or network traffic monitoring: Note: Although Telegram's pps is low, when the network is poor, MTPROTO may perform more aggressive retransmissions, potentially doubling pps for a short period. Also, enabling "self-destructing messages" or "secret chat" mode slightly increases the number of handshake packets. Would you like me to help you compare the data packet differences between Telegram's "normal chat" and "end-to-end encrypted secret chat"?>

4. Google Gemini (Feb 1, 2026 version, Gemini 3)

Query: <What is an RCC state machine?>

Output: <In mobile communications (e.g., LTE, 5G) and smartphone power consumption research, what you mentioned is likely RRC (Radio Resource Control, Radio Resource Control) state machine. It is one of the core mechanisms determining mobile phone power consumption. Simply put, the RRC state machine defines the connection states between the mobile phone's radio (Modem/antenna) and the base station. Different states correspond to different power consumption levels.

1. Why is it important for power consumption research? The 4G/5G module of a phone is a major power consumer. To save power, the radio cannot remain in "full-power cruise" mode all the time. The RRC state machine acts like an automatic energy-saving switch: When there is data to send: Switch to a high-power state. When there is no data to send: Wait for a period, then automatically downgrade to a low-power or sleep state.

2. Three Core States (Taking LTE as an example) State Name Description Power Level RRC_IDLE (Idle) The phone is not connected to the base station, only listening to "broadcasts." Extremely low (close to 0) RRC_CONNECTED (Connected) Actively transmitting data, antenna fully on. Very high (can reach hundreds of milliwatts) RRC_IDLE with DRX (Discontinuous Reception) An intermediate state, waking up periodically to check for calls. Medium to low Note: In 5G, an RRC_INACTIVE (Inactive) state is added, designed to allow extremely fast connection recovery while maintaining low power consumption similar to the idle state.

3. Key Concept in Power Consumption Research: Tail Energy This is a term you will often encounter in papers. After Telegram or Facebook finishes sending those few text packets (instant pps burst), the RRC state machine does not immediately jump back to the

IDLE state. Instead, it remains in the high-energy-consumption state for several seconds (typically 5-10 seconds) in case you want to send another message immediately. The pain point: Even if you only send 1KB of text, due to this "tail time," the phone may consume enough power to sustain a high-energy connection for 10 seconds. Impact of pps: If your application sends a packet every few seconds (heartbeat packets), the RRC state machine will be constantly "awakened" and unable to enter sleep, leading to rapid battery drain. This is why optimizing pps and heartbeat frequency is crucial for mobile battery life.

4. Summary The RRC state machine is the logic controlling the mobile phone radio's "when to be excited and when to be calm." In smartphone power consumption research, understanding state transition thresholds and timers is more important than simply looking at packet sizes. Would you like me to show you the energy-saving improvements made in RRC state transitions in 5G compared to 4G?>

5. Google Gemini (Feb 1, 2026 version, Gemini 3)

Query: <Search for the information of the CALCE battery dataset>

Output: <The CALCE (Center for Advanced Life Cycle Engineering, University of Maryland) battery dataset is one of the most authoritative public data sources in the global lithium-ion battery research field, especially for state estimation, life prediction, and reliability analysis.

Here is a summary of the core information about this dataset:

1. Dataset Overview The CALCE dataset covers various battery formats and chemical systems, primarily used to support research on State of Charge (SOC) estimation, State of Health (SOH) monitoring, and Remaining Useful Life (RUL) prediction.

2. Main Battery Models and Types The dataset is divided into three major categories by battery format: Cylindrical Cells: INR 18650-20R: NMC battery (LiNiMnCo/Graphite), capacity 2000 mAh. Includes low-current OCV tests at different temperatures (0°C, 25°C, 45°C), incremental OCV tests, and various dynamic stress tests (DST, FUDS, US06, BJDST). A123: LiFePO₄ battery, capacity 1110 mAh. Includes dynamic tests and OCV tests within the range of -10°C to 50°C. Prismatic Cells: CS2 Series: LCO battery (LiCoO₂), capacity 1100 mAh. Focuses on long-term cycle life testing, including tests at various discharge rates (0.5C, 1C) and with random current fluctuations. CX2 Series: LCO battery (LiCoO₂), capacity 1350 mAh. Includes pulse discharge loads, cycles with temperature switching, etc. Pouch Cells: PL Series: LCO (Graphite/LiCoO₂), capacity 1500 mAh. Includes cycle tests (studying the impact of different SOC ranges and discharge rates on lifespan) and storage tests (studying capacity fade during storage at different temperatures and SOC levels).

3. Test Items and Content The dataset includes a very comprehensive range of test types, directly usable for parameter identification of equivalent circuit models (like the second-order RC model you mentioned): Cycling Tests: Full charge/discharge cycles, partial charge/discharge cycles. Dynamic Stress Tests: Tests simulating real driving stress in electric vehicles, such as DST (Dynamic Stress Test), FUDS (Federal Urban Driving Schedule). OCV Tests: Provide precise curves of open-circuit voltage versus SOC. Stor-

age Tests: Investigate calendar life at different ambient temperatures (-40°C to 50°C). Impedance Measurements: Provide electrochemical impedance spectroscopy data.

4. Data Format and Access Format: Data is typically provided in Excel (.xlsx) or text (.txt) format, facilitating import into MATLAB, Python, or Python data processing libraries (like Pandas) for algorithm validation. Usage: Many top papers on Kalman Filters (EKF/UKF) or deep learning for SOC prediction use this dataset as a benchmark. Citation Requirements: When using this data for publication, relevant CALCE research papers must be cited (detailed citation lists are available at the bottom of the website pages).

Summary and Suggestion: If you want to validate the application of the second-order RC equivalent circuit model on smartphone batteries, the CS2 or CX2 series (prismatic LCO batteries) are most chemically similar to modern smartphone batteries. If you focus on SOC prediction under dynamic loads, the rich dynamic test data provided by the INR 18650-20R is an excellent choice.>

6.Google Gemini (Feb 1, 2026 version, Gemini 3)

Query: <Search for papers related to lithium battery dynamics models>

Output: <Research papers on lithium-ion battery dynamic models are extensive and are typically categorized into three major types based on model complexity, physical meaning, and computational requirements: Equivalent Circuit Models (ECM), Electrochemical Models (EM), and Black-Box/Machine Learning Models.

Here is a detailed overview of representative papers and research directions for these core dynamic model categories:

1. Equivalent Circuit Models (ECM) These models are most widely used in Battery Management Systems (BMS) for phones and electric vehicles, balancing computational load and accuracy. Classic Second-Order RC Model (Extension of Thevenin model): Paper: Stroe, et al. "Lithium-ion battery dynamic model for wide range of operating conditions." (2017) Core: Proposes a second-order RC model and uses current pulse techniques (HPPC) for parameter identification under different temperatures, loads, and SOC levels, validating its high accuracy in simulating dynamic behavior. Comparative Study of Models: Paper: He, Hongwen, et al. "Evaluation of Lithium-Ion Battery Equivalent Circuit Models for State of Charge Estimation by an Experimental Approach." (2011) Core: Compares models like Rint, RC, Thevenin, and PNGV, concluding that the Dual Polarization (DP) model (i.e., second-order RC) performs best in simulating electrochemical polarization and concentration polarization. Nonlinearity and Parameter Identification: Paper: Mitsubishi Electric Research Laboratories. "On Parameter Identification of an Equivalent Circuit Model for Lithium-Ion Batteries." (2017) Core: Discusses how to identify nonlinear parameters through small-current experiments to improve model robustness across the entire lifespan.

2. Electrochemical Dynamic Models (Electrochemical Models) These models are based on microscopic physical processes (diffusion, kinetic reactions), offering the strongest physical meaning, often used for life analysis and safety studies. P2D Model (Pseudo-

2D Model): Foundational Paper: Doyle, M., and Newman, J. "Modeling of Galvanostatic Charge and Discharge of the Lithium/Polymer/Lithium System." (Foundational Work) Modern Developments: Many papers study how to simplify this model for real-time operation (e.g., Single Particle Model SPM), such as MDPI "Modelling and Estimation in Lithium-Ion Batteries: A Literature Review" (2023). Electrochemical-Thermal Coupled Models: Paper: MDPI "Electrochemical–Thermal Modeling of Lithium-Ion Batteries: An Analysis of Thermal Runaway with Observation on Aging Effects" (2025) Core: Investigates the impact of aging on thermal dynamics, revealing the opposing effects of increasing internal resistance and capacity fade on thermal runaway risk.

3. Advanced Dynamic Behavior Modeling (Advanced & Hybrid Models) Mechanical Analogy Dynamic Models: Paper: Alsharif, K. I. "Dynamic Modeling and Thermal Characterization of Lithium-Ion Batteries." Core: Proposes an innovative "mass-spring-damper" system analogy model, using mechanical vibration theory to simulate battery terminal voltage fluctuations and mass transfer effects. Fractional-Order Models: These models use fractional calculus to describe internal diffusion phenomena, offering more precise descriptions of low-frequency impedance characteristics than traditional RC models.

4. Recommended Review Literature (Essential Reading for Beginners) If you need a comprehensive overview, consider the following recent reviews: *Modelling and Estimation in Lithium-Ion Batteries: A Literature Review* (2023): Covers the full spectrum from complex physical models to simplified real-time estimation algorithms. *Thermal Behavior Modeling of Lithium-Ion Batteries: A Comprehensive Review*: Focuses on thermal effects in dynamic models and their impact on lifespan.

Suggested Search Resources You can directly search for keywords like *Lithium-ion battery dynamic model* or *Parameter identification of battery ECM* in the following databases: IEEE Xplore / ScienceDirect, MDPI Energies / Applied Energy, CALCE (Center for Advanced Life Cycle Engineering) official associated papers. Would you like me to break down the specific formulas (e.g., differential equations of the second-order RC model) or parameter identification methods from any particular paper for you?>